



ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC KINH TẾ - LUẬT

VIETNAM NATIONAL UNIVERSITY HO CHI MINH CITY
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DATA MINING

Lecture 3: Mining Descriptive Statistics and Analytical Characterization with Data



Lecture 3 outline

Section	What we cover
3.1.	Data classification: objects, attributes, and the four attribute types
3.2.	Descriptive Statistics Measure: central tendency, dispersion, five-number summary
3.3.	Data visualization: histogram, boxplot, quantile plot, scatter, and beyond two dimensions
3.4.	Measuring data similarities and differences: distance and similarity measures
Lab.	Lab 3 in Google Colab on the three Saigon Bean files

Lecture 3 roadmap and outcome

1 Classify any column of a real table by attribute type, and say what that type permits you to compute

CLO2

2 Compute and interpret central tendency, dispersion and the five-number summary, then read the skew

CLO2

3 Choose the right chart for a question, and state what each chart hides as well as what it shows

CLO2

4 Measure how similar two objects are, and defend the measure you chose against an alternative

CLO4

Dataset Description

File	Rows	One row is	Columns
saigon_bean_stores.csv	12	One store	store_id, store_name, district, store_type, seats
saigon_bean_products.csv	17	One product on the menu	product_id, product_name, category, list_price
saigon_bean_transactions.csv	97,239	One line of one receipt	order_id, store_id, order_time, channel, customer_id, product_id, quantity, unit_price, discount, line_total

What it covers

- 12 stores in Ho Chi Minh City, four types: mall, campus, residential, transit
- 91 days of trading, 1 June to 31 August 2026
- 51,108 orders, 97,239 order lines, 4,777 identified customers

Why three files and not one

- This is how a real transaction system stores it
- Stores and products are dimensions, transactions are the facts
- Joining them is your first task in Lab 2

3.1

Data Classification

Data objects, attributes, and the four attribute types

Types of Data Sets

Objects and attributes

Data object

Record

- Relational records
- Data matrix, e.g., numerical matrix, crosstabs
- Document data: text documents: term-frequency vector
- Transaction data

Graph and network

- World Wide Web
- Social or information networks
- Molecular Structures

Ordered

- Video data: sequence of images
- Temporal data: time-series
- Sequential Data: transaction sequences

Genetic sequence data

- Spatial, image and multimedia
- Spatial data
- Image data
- Video data

	team	coach	play	ball	score	game	win	lost	timeout	season
Document 1	3	0	5	0	2	6	0	2	0	2
Document 2	0	7	0	2	1	0	0	3	0	0
Document 3	0	1	0	0	1	2	2	0	3	0

<i>TID</i>	<i>Items</i>
1	Bread, Coke, Milk
2	Beer, Bread
3	Beer, Coke, Diaper, Milk
4	Beer, Bread, Diaper, Milk
5	Coke, Diaper, Milk

Important Characteristics of Structured Data

- Dimensionality
 - Curse of dimensionality
- Sparsity
 - Only presence counts
- Resolution
 - Patterns depend on the scale
- Distribution
 - Centrality and dispersion

3.1 Data Classification

Objects and attributes

Data object

- A data set is made up of data objects
- A data object represents one entity
- In our files: one store, one product, one line of a receipt
- Also called *a sample, an example, an instance, a data point, a tuple, a record*
- In a table, a row is an object

Attribute

- A data field representing one characteristic of an object
- Also called *a dimension, a feature or a variable*
- Econometrics calls it a variable, machine learning calls it a feature, and databases call it a column
- In a table, a column is an attribute

store_id	store_name	district	store_type	seats	← attributes (columns)
1	Nguyen Hue	District 1	mall	70	← one object (row)
7	University Village	Thu Duc	campus	40	
11	Tan Son Nhat T1	Tan Binh	transit	24	

Everything in this chapter depends on reading a table in these two directions at once: down a column to describe an attribute, across a row to describe an object.

3.1 Data Classification

The four attribute types

Nominal	Categories, states, or names of things. No order at all.	<code>district, channel, category, store_name</code>
Binary	A nominal attribute with exactly two states, 0 and 1.	<code>is_mall, has_loyalty_card, refunded</code>
Ordinal	Values have a meaningful order, but the gap between consecutive values is unknown.	<code>size = {small, medium, large}, ratings</code>
Numeric	A measured quantity, integer or real valued. Splits into interval and ratio.	<code>seats, quantity, unit_price, line_total</code>

Numbers in a column do not make it numeric. `store_id` is stored as an integer and it is nominal, because the average of two store identifiers means nothing.

3.1 Data Classification

Nominal and binary attributes

Nominal

- Values are labels. There is no order, no distance, no arithmetic
- Permitted: counting, mode, and equality tests only
- In our data: district with 7 values, store_type with 4, category with 4, channel with 3

ASYMMETRIC, AND WHY IT MATTERS HERE

Build an attribute bought_merchandise for each receipt. Only 1.4% of revenue is merchandise, so a 1 is far more informative than a 0. Two receipts that both lack merchandise share nothing in particular. Section 2.4 gives this case its own similarity measure.

Binary

- A nominal attribute with only two states, written 0 and 1
- Symmetric: both outcomes matter equally, for example gender
- Asymmetric: the two outcomes are not equally informative, for example a positive medical test

A FAULT ALREADY IN THE FILE

Channel should have three values. The raw file has six: in-store, app, delivery, and also In-Store, App, Delivery. Thirty rows carry the capitalized spelling. A machine sees six categories where a human sees three.

3.1 Data Classification

Ordinal attributes

Order without a known magnitude

- The values can be ranked, and the gap between consecutive ranks is unknown
- Permitted: mode, median, and rank-based statistics
- The mean is not permitted, because the distance between ranks is undefined

```
size      = {small, medium, large}
rating    = {1, 2, 3, 4, 5 stars}
grade     = {A+, A, B+, B, C, D+, D, F}
queue_band = {short, normal, long}

seats_band = {small, medium, large}
             from seats: 16-30, 31-50, 51-70
```

THE TRAP THAT CATCHES EVERYONE

A five-star rating is ordinal. The average rating that every platform reports treats it as numeric, which assumes the step from 1 to 2 equals the step from 4 to 5. That assumption is convenient and it is not justified by the measurement. Report the median and the full distribution alongside the mean.

Section 2.4 shows the standard repair: map the ranks to $[0, 1]$ first, then treat the result as numeric.

3.1 Data Classification

Numeric attributes: interval and ratio, discrete and continuous

Interval-scaled

- Measured on a scale of equal-sized units, and values have order
- No true zero point, so ratios are meaningless
- order_time, calendar dates, temperature in Celsius
- You may subtract two order times. You may not say one is twice the other

Ratio-scaled

- An inherent zero point exists, so ratios carry meaning
- seats, quantity, unit_price, line_total, revenue
- A 70-seat store really does have twice the seating of a 35-seat store
- Most money and count variables in economics are ratio-scaled

Discrete

- A finite or countably infinite set of values
- Sometimes, represented as integer variables
- Note: Binary attributes are a special case of discrete attributes

Continuous

- Real-valued, stored as floating point and measured to a finite precision
- Practically, real values can only be measured and represented using a finite number of digits
- Continuous attributes are typically represented as floating-point variables

Interval against ratio decides whether a ratio is meaningful. Discrete against continuous decides which chart and which model are available.

3.1 Data Classification

Classify every column of `saigon_bean_transactions.csv`

Attribute	Type	What is in the file	Why
order_id	Nominal	51,108 distinct values	A number used as a label
store_id	Nominal	12 distinct values	A key, not a quantity
order_time	Numeric, interval	1 Jun to 31 Aug 2026	Differences are meaningful, there is no true zero
channel	Nominal	3 real values, 6 spellings	Unordered categories
customer_id	Nominal	4,777 distinct, 34.5% empty	A label. Empty means a walk-in customer
product_id	Nominal	17 distinct values	A key into the product table
quantity	Numeric, ratio, discrete	1 and 2, plus 25 negative rows	A count, and zero means none
unit_price	Numeric, ratio, continuous	25,000 to 5,500,000 VND	Money, with a true zero
discount	Numeric, ratio, continuous	zero on 93.4% of lines	Money, with a true zero
line_total	Numeric, ratio, continuous	-118,000 to 11,000,000 VND	Derived from quantity, price and discount

3.1 Data Classification

What each type lets you compute

Operation	Nominal	Binary	Ordinal	Interval	Ratio
Count, frequency, mode	yes	yes	yes	yes	yes
Median and quantiles	no	no	yes	yes	yes
Add and subtract, mean	no	no	no	yes	yes
Multiply, divide, ratios	no	no	no	no	yes
Standard deviation	no	no	no	yes	yes

Practice 1

- 1 Classify every column of `saigon_bean_stores.csv` and `saigon_bean_products.csv` by attribute type and mark each numeric column as discrete or continuous.
- 2 For each of the five faults on the previous slide, say what you would do about it and what the cost of doing nothing would be. One line each, written for a manager rather than for a programr.
- 3 A colleague computes the mean of `store_id` and reports 6.5 as the average store. Explain in two sentences why the number is arithmetically correct and analytically empty.

3.2

Descriptive Statistics Measure

Central tendency, dispersion, and the shape of a distribution

Descriptive Statistics Measure

- Motivation
 - To better understand the data: central tendency, variation and spread
- Data dispersion characteristics
 - median, max, min, quantiles, outliers, variance, etc.
- Numerical dimensions correspond to sorted intervals
 - Data dispersion: analyzed with multiple granularities of precision
 - Boxplot or quantile analysis on sorted intervals
- Dispersion analysis on computed measures
 - Folding measures into numerical dimensions
 - Boxplot or quantile analysis on the transformed cube

3.2 Descriptive Statistics Measure

Measuring the central tendency

Mean

- The arithmetic average of all values
- Weighted mean when objects carry different importance
- Trimmed mean drops the extreme tails before averaging
- Sensitive to every extreme value

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \mu = \frac{\sum x}{N}$$

$$\bar{x} = \frac{\sum_{i=1}^n w_i x_i}{\sum_{i=1}^n w_i}$$

Note: n is sample size and N is population size.

Median

- The middle value when the data is sorted
- The average of the two middle values when the count is even
- Estimated by interpolation within a class for grouped data
- Unaffected by how extreme the extremes are

$$median = L_1 + \left(\frac{n/2 - (\sum freq)l}{freq_{median}} \right) width$$

Mode

- The value that occurs most often
- The only central measure available for a nominal attribute
- A distribution can be unimodal, bimodal or multimodal
- Empirical formula:

$$mean - mode = 3 \times (mean - median)$$

3.2 Descriptive Statistics Measure

Example 1. Basic calculation

Student	Study hours
S1	2
S2	3
S3	4
S4	5
S5	5
S6	6
S7	8
S8	8
S9	8
S10	10

1. By hand

- Mean
- Median
- Mode
- Standard Deviation

2. Coding

```
# Mean
mean = np.mean(study_hours)
# Median
median = np.median(study_hours)
# Mode
mode_value = mode(study_hours)
# Standard Deviation
std = np.std(study_hours)
```

3.2 Descriptive Statistics Measure

The three measures on the Saigon Bean basket

Measure	Order line (n = 97,214)	Basket per order (n = 51,106)	Reading
Mean	57,226 VND	108,856 VND	Pulled up by a few enormous values
Median	52,000 VND	97,000 VND	The typical customer
Mode	45,000 VND	45,000 VND	One iced milk coffee, nothing else
Std. deviation	78,960 VND	116,109 VND	Larger than the mean itself

READ THE THREE TOGETHER

Mode 45,000, median 97,000, mean 108,856. The three separate in one direction, and the order mode < median < mean is the signature of a right-skewed distribution. The mean sits 11,856 dong above the median.

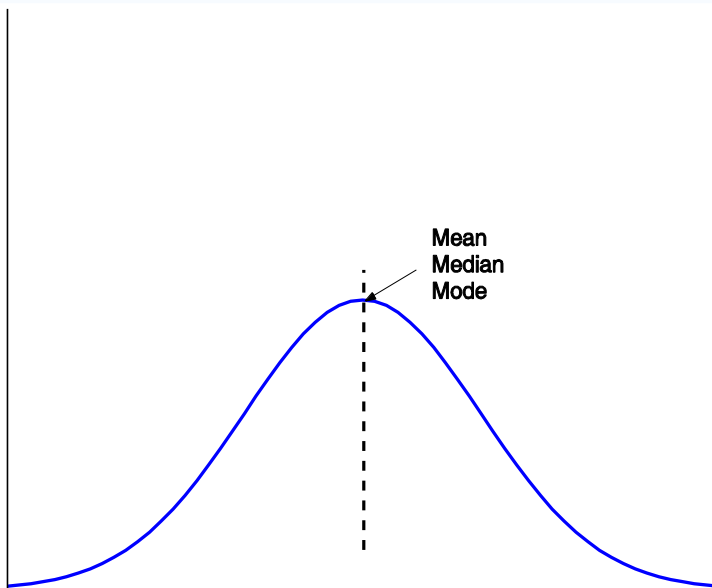
A STANDARD DEVIATION LARGER THAN THE MEAN

That is a warning, not a result. It usually means the tail contains values that do not belong to the same process. Here it is the twelve price-scale errors, one of which turns a single cheesecake into an 11 million dong basket.

3.2 Descriptive Statistics Measure

Symmetric and skewed data

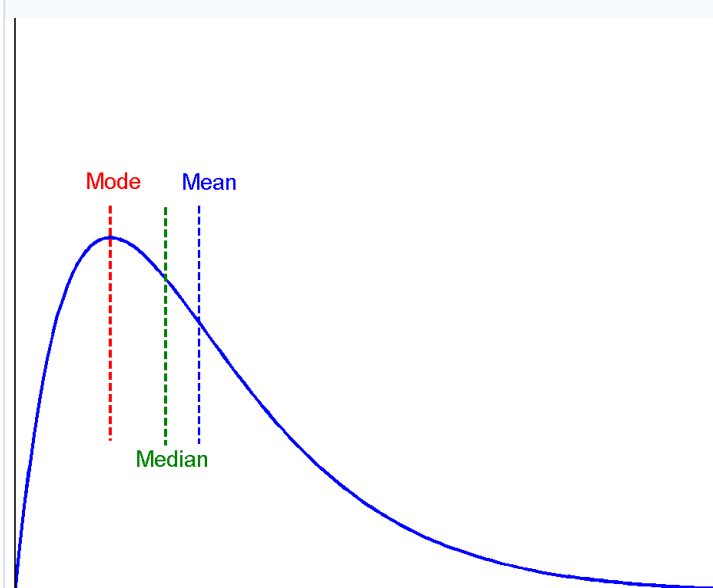
Symmetric



$$\text{mean} = \text{median} = \text{mode}$$

Heights, measurement error, a sum of many small independent effects

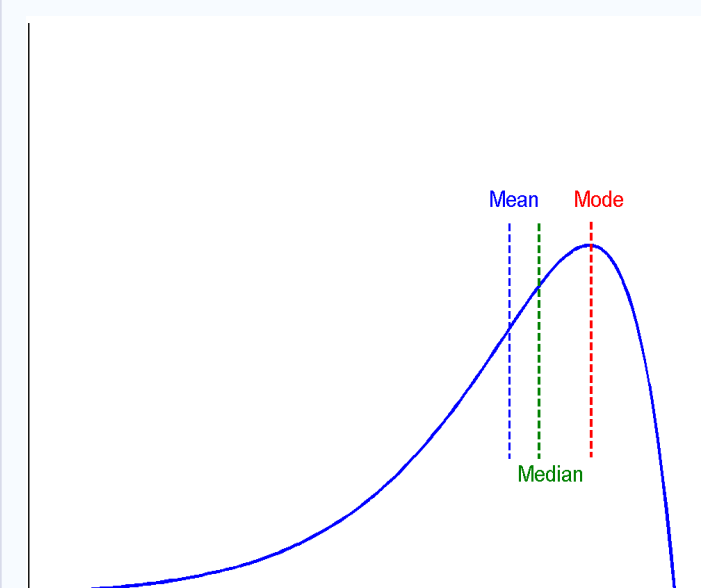
Positively skewed



$$\text{mode} < \text{median} < \text{mean}$$

Basket value, income, firm size, city population. The long tail is on the right

Negatively skewed



$$\text{mean} < \text{median} < \text{mode}$$

Exam marks with a ceiling, age at retirement. The long tail is on the left

3.2 Descriptive Statistics Measure

Measuring the Dispersion of Data

Quartiles, outliers and boxplots

- **Quartiles:** Q_1 (25th percentile), Q_3 (75th percentile)
- **Inter-quartile range:** $IQR = Q_3 - Q_1$
- **Five number summary:** min, Q_1 , median, Q_3 , max
- **Boxplot:** ends of the box are the quartiles; median is marked; add whiskers, and plot outliers individually
- **Outlier:** usually, a value higher/lower than $1.5 \times IQR$

Variance and standard deviation (sample: s , population: σ)

- **Variance:** (algebraic, scalable computation)

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \left[\sum_{i=1}^n x_i^2 - \frac{1}{n} \left(\sum_{i=1}^n x_i \right)^2 \right]$$

$$\sigma^2 = \frac{1}{N} \sum_{i=1}^n (x_i - \mu)^2 = \frac{1}{N} \sum_{i=1}^n x_i^2 - \mu^2$$

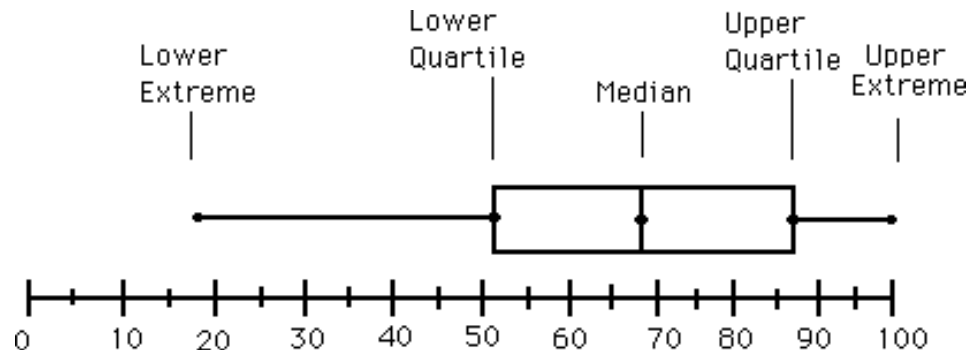
- **Standard deviation** s (or σ) is the square root of variance s^2 (or σ^2)

2.2 Basic Statistics of the Data

Boxplot Analysis

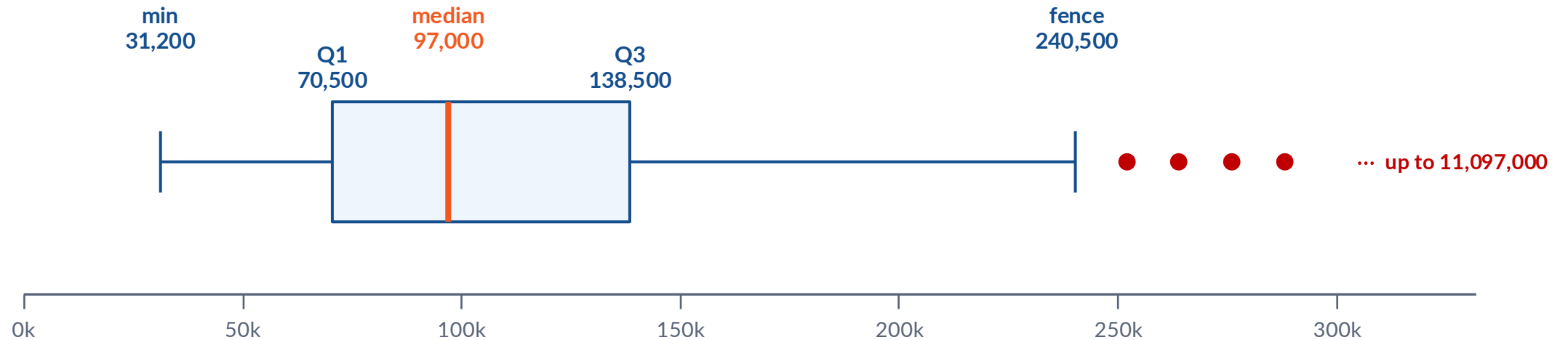
Boxplots Analysis

- **Five-number summary** of a distribution
 - Minimum, Q1, Median, Q3, Maximum
- **Boxplot**
 - Data is represented with a box
 - The ends of the box are at the first and third quartiles, i.e., the height of the box is IQR
 - The median is marked by a line within the box
 - Whiskers: two lines outside the box extended to Minimum and Maximum
 - Outliers: points beyond a specified outlier threshold, plotted individually



3.2 Descriptive Statistics Measure

Boxplot of the Saigon Bean basket value



HOW THE FENCE IS BUILT

$IQR = 138,500 - 70,500 = 68,000$. Upper fence = $Q3 + 1.5 \times IQR = 240,500$. Eight hundred and ninety-three orders sit above it, which is 1.7% of all orders.

NOT EVERY OUTLIER IS AN ERROR

Most of the 893 are genuine large group orders. Twelve of them are the price-scale fault from section 2.1. The boxplot finds candidates. Only the product table can tell you which candidate is a mistake.

Half of all Saigon Bean orders fall between 70,500 and 138,500 dong. That single sentence is worth more to a manager than the mean.

3.2 Descriptive Statistics Measure

The normal curve, and when it does not apply

Properties worth memorizing

- The curve is symmetric, and mean, median and mode coincide
- About 68% of values lie within one standard deviation of the mean
- About 95% lie within two, and about 99.7% within three
- Two parameters, the mean and the standard deviation, describe it completely

Why it matters here

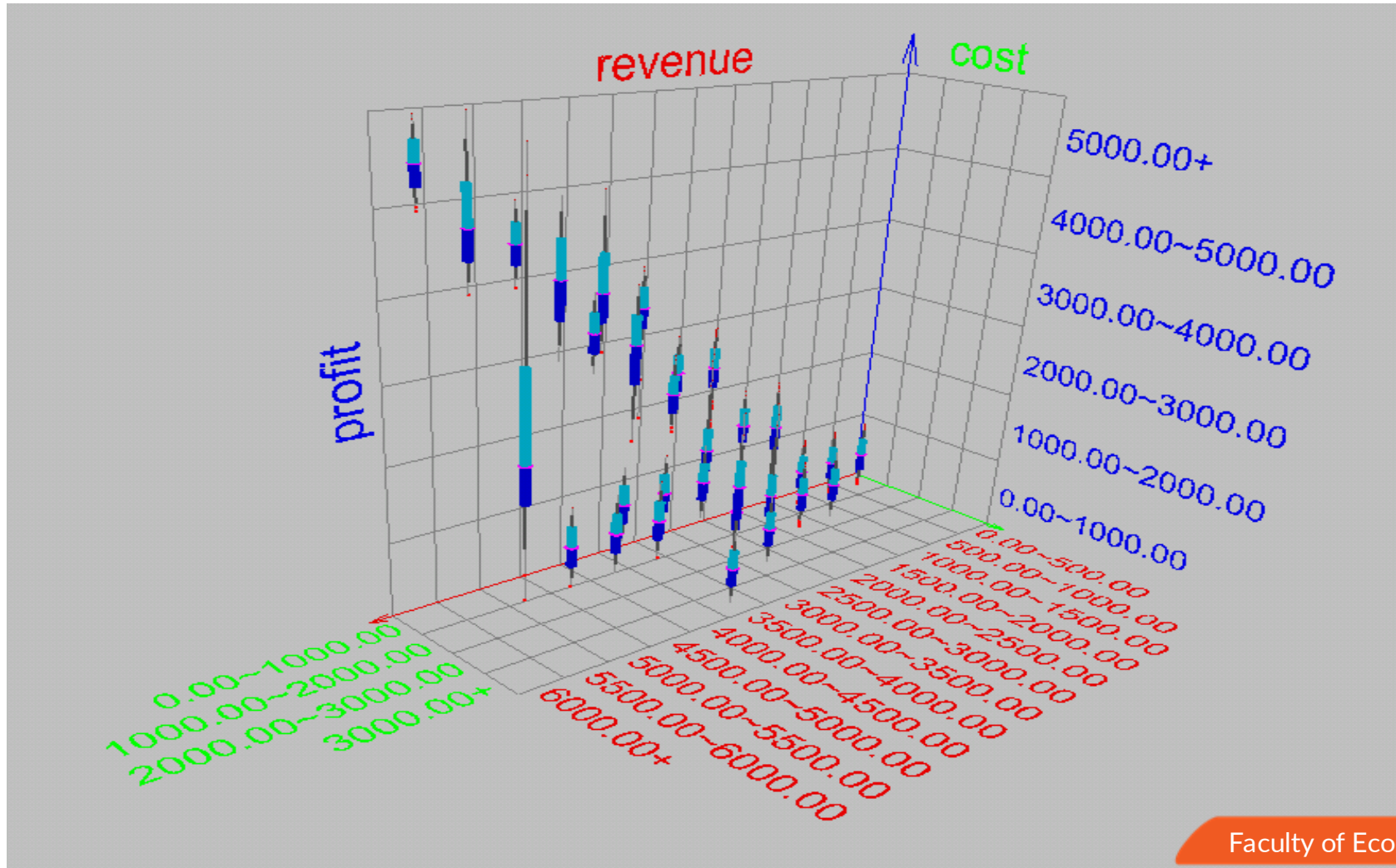
- Many classical tests assume it, including most of what you met in econometrics
- The three-sigma rule is a common outlier criterion
- Standardizing an attribute, the z-score, is built on it

THE SAIGON BEAN BASKET IS NOT NORMAL

Mean 108,856 and standard deviation 116,109. One standard deviation below the mean gives a negative basket value, which cannot happen. The minimum is 31,200. When the three-sigma rule produces an impossible bound, the assumption behind it has already failed, and the quantile measures are the ones to trust.

3.2 Descriptive Statistics Measure

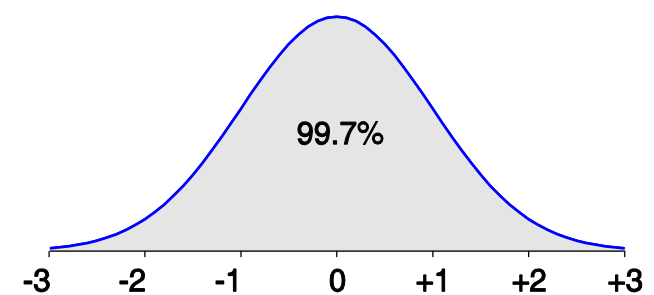
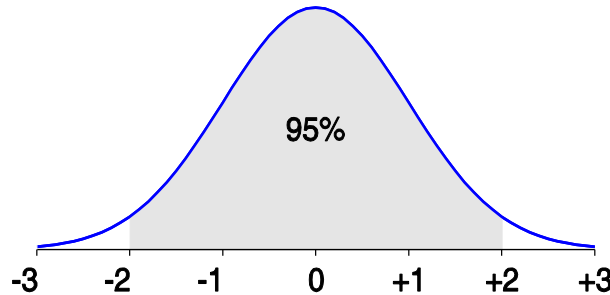
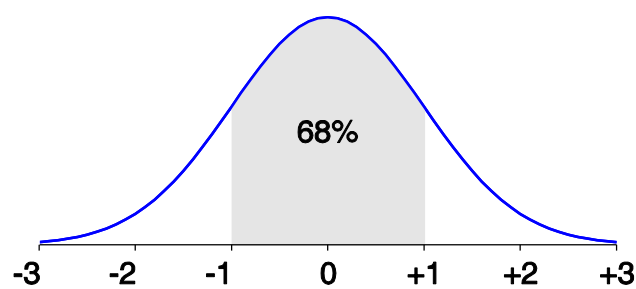
Visualization of Data Dispersion: 3-D Boxplots



3.2 Descriptive Statistics Measure

Properties of Normal Distribution Curve

- The normal (distribution) curve
 - From $\mu - \sigma$ to $\mu + \sigma$: contains about 68% of the measurements (μ : mean, σ : standard deviation)
 - From $\mu - 2\sigma$ to $\mu + 2\sigma$: contains about 95% of it
 - From $\mu - 3\sigma$ to $\mu + 3\sigma$: contains about 99.7% of it



3.2 Descriptive Statistics Measure

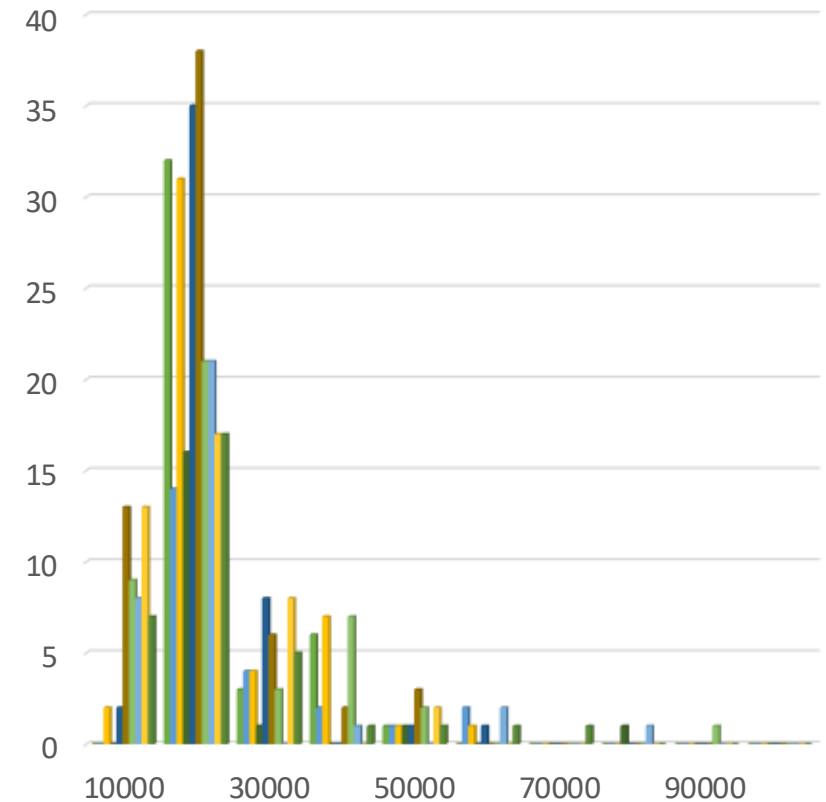
Graphic Displays of Basic Statistical Descriptions

- **Boxplot:** graphic display of five-number summary
- **Histogram:** x-axis are values, y-axis repres. frequencies
- **Quantile plot:** each value x_i is paired with f_i indicating that approximately $100 f_i \%$ of data are $\leq x_i$
- **Quantile-quantile (q-q) plot:** graphs the quantiles of one univariant distribution against the corresponding quantiles of another
- **Scatter plot:** each pair of values is a pair of coordinates and plotted as points in the plane

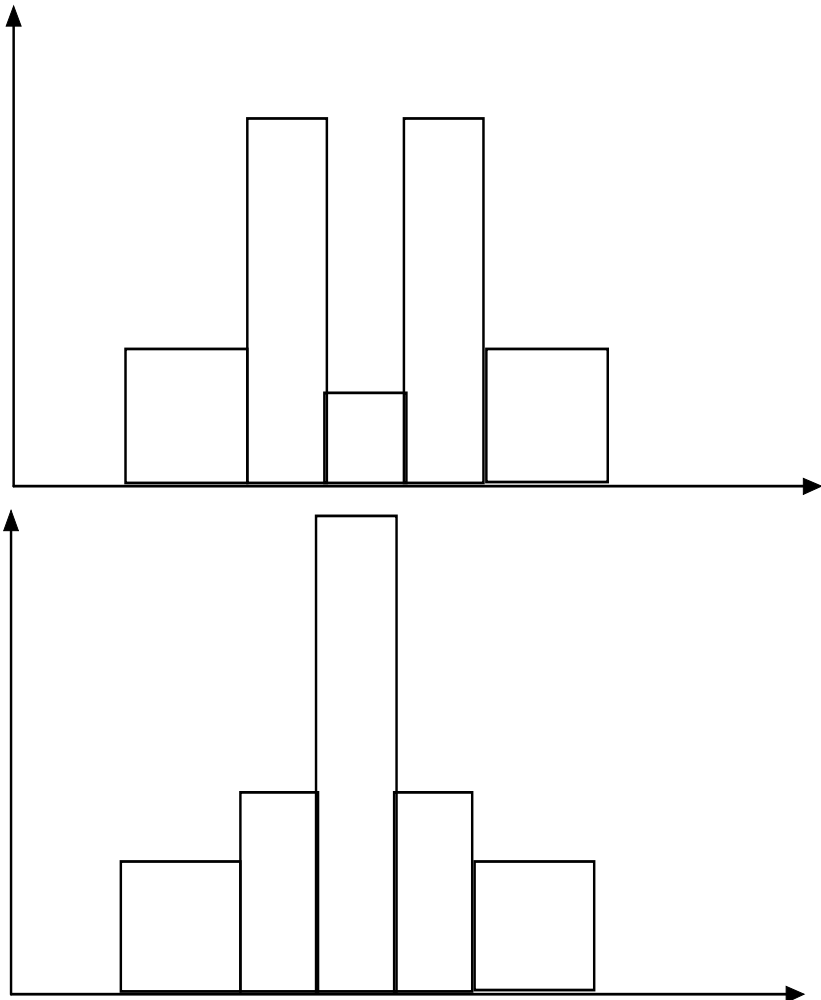
3.2 Descriptive Statistics Measure

Histogram Analysis

- Histogram: Graph display of tabulated frequencies, shown as bars
- It shows what proportion of cases fall into each of several categories
- Differs from a bar chart in that it is the *area* of the bar that denotes the value, not the height as in bar charts, a crucial distinction when the categories are not of uniform width
- The categories are usually specified as non-overlapping intervals of some variable. The categories (bars) must be adjacent



3.2 Descriptive Statistics Measure



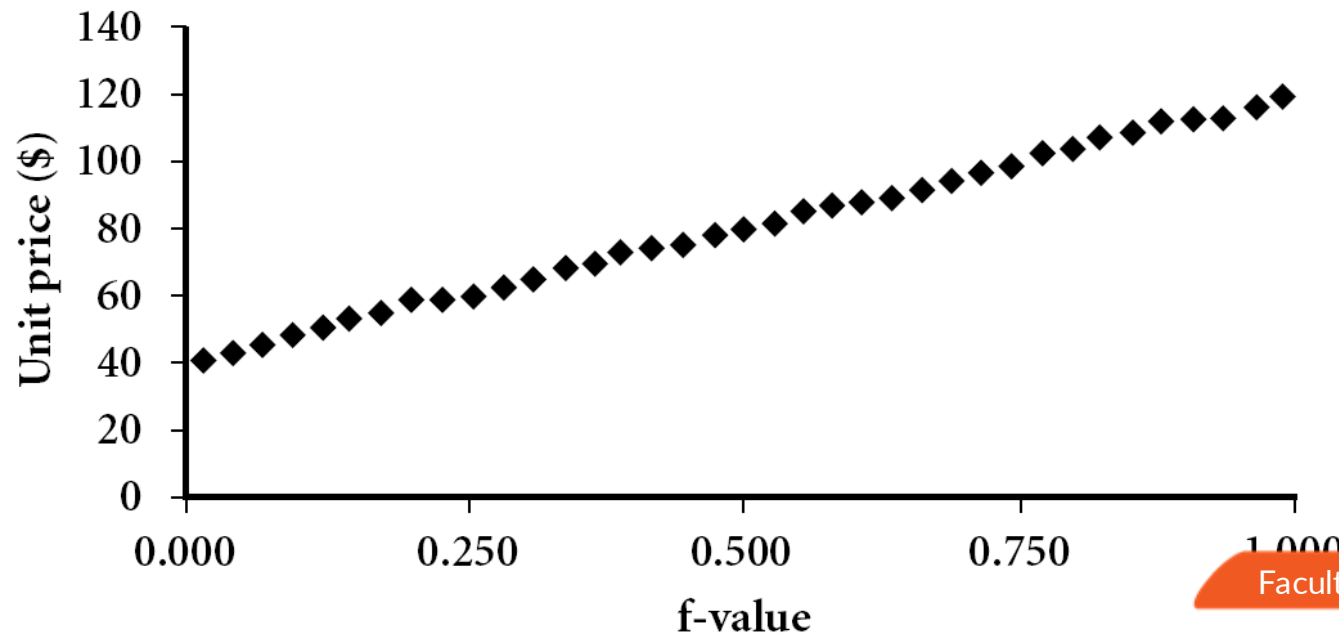
Histograms Often Tell More than Boxplots

- The two histograms shown in the left may have the same boxplot representation
 - The same values for: min, Q1, median, Q3, max
- But they have rather different data distributions

3.2 Descriptive Statistics Measure

Quantile Plot

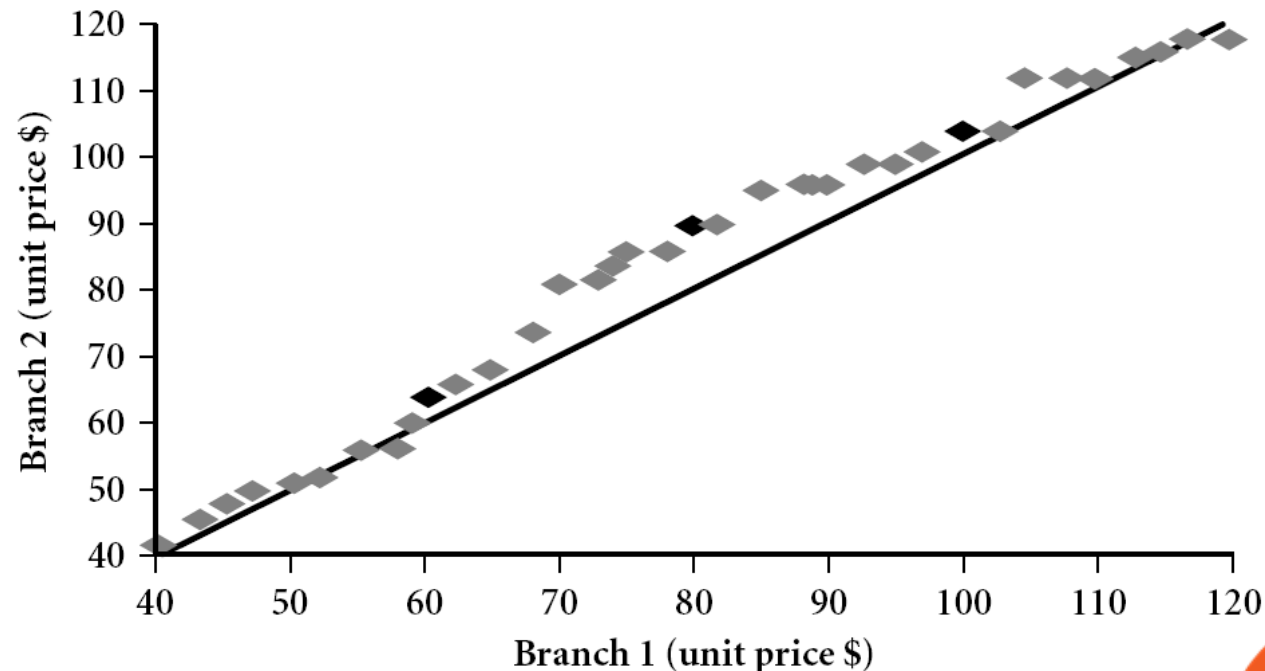
- Displays all of the data (allowing the user to assess both the overall behavior and unusual occurrences)
- Plots **quantile** information
 - For a data x_i data sorted in increasing order, f_i indicates that approximately 100 $f_i\%$ of the data are below or equal to the value x_i



3.2 Descriptive Statistics Measure

Quantile-Quantile (Q-Q) Plot

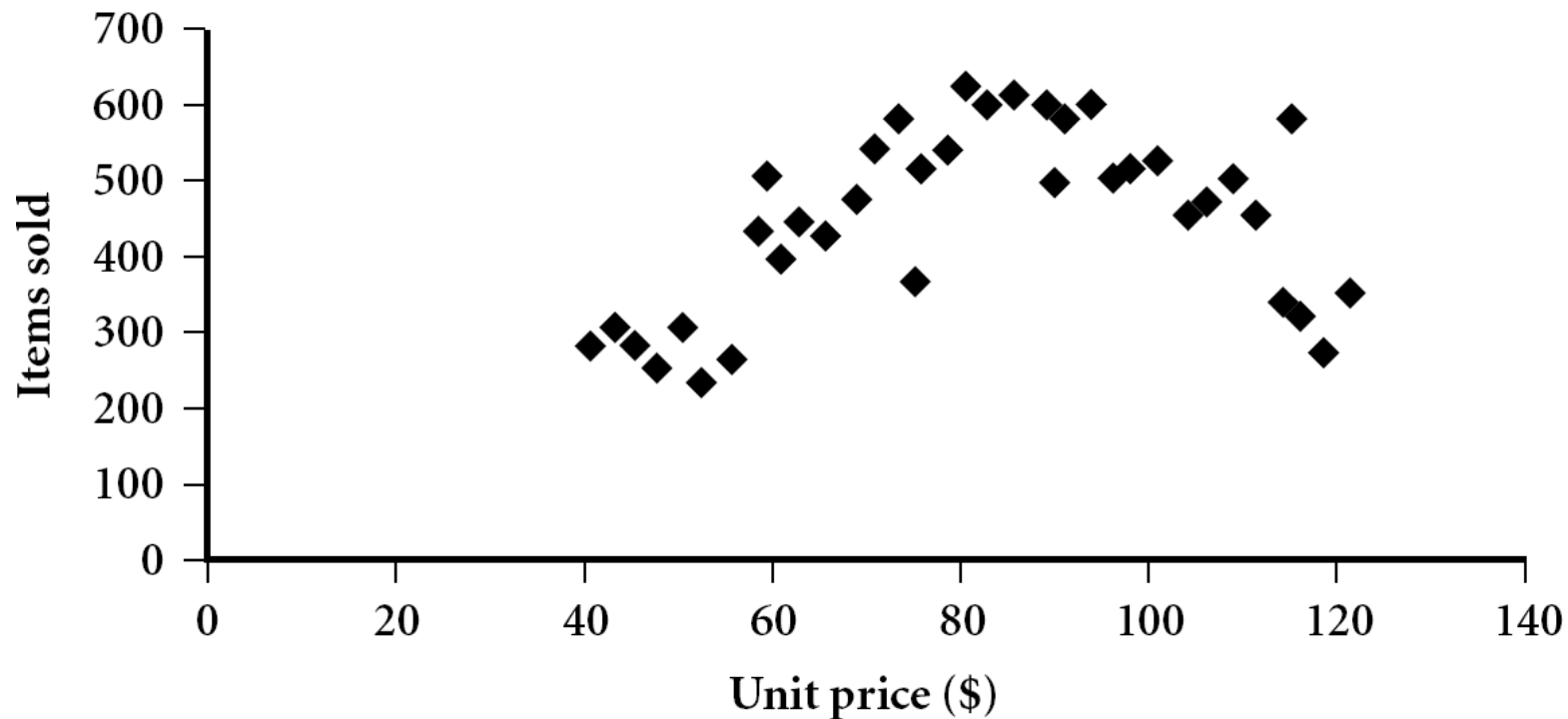
- Graphs the quantiles of one univariate distribution against the corresponding quantiles of another
- View: Is there is a shift in going from one distribution to another?
- Example shows unit price of items sold at Branch 1 vs. Branch 2 for each quantile. Unit prices of items sold at Branch 1 tend to be lower than those at Branch 2.



3.2 Descriptive Statistics Measure

Scatter plot

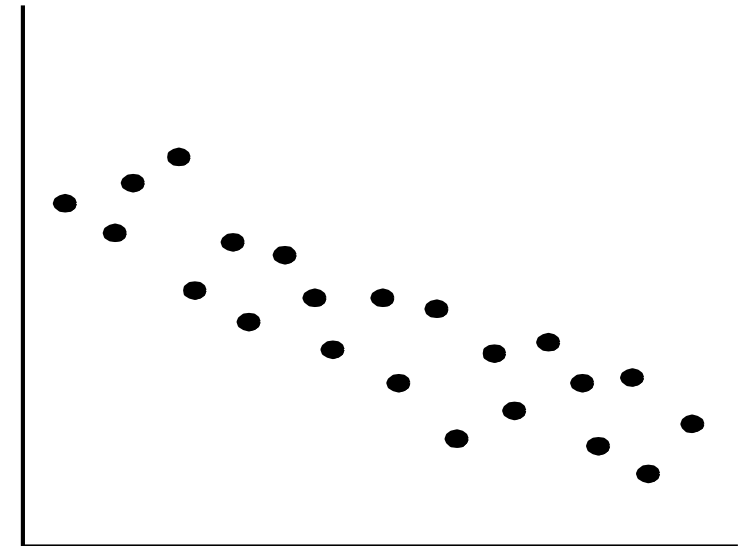
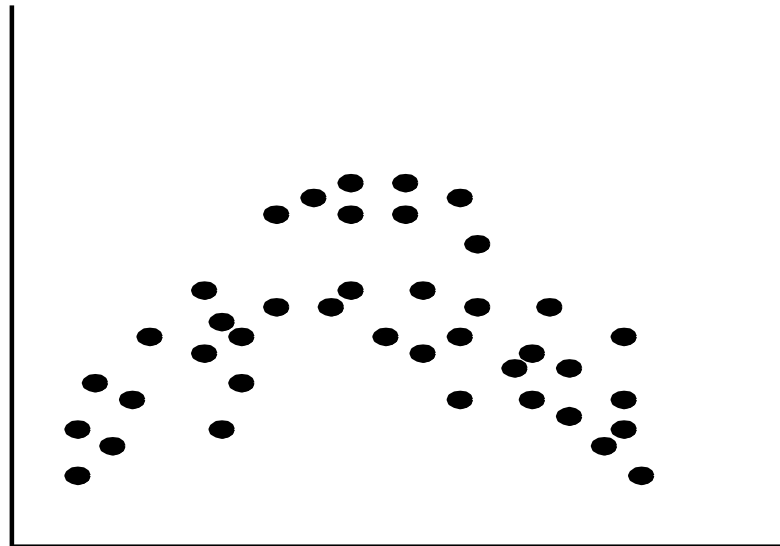
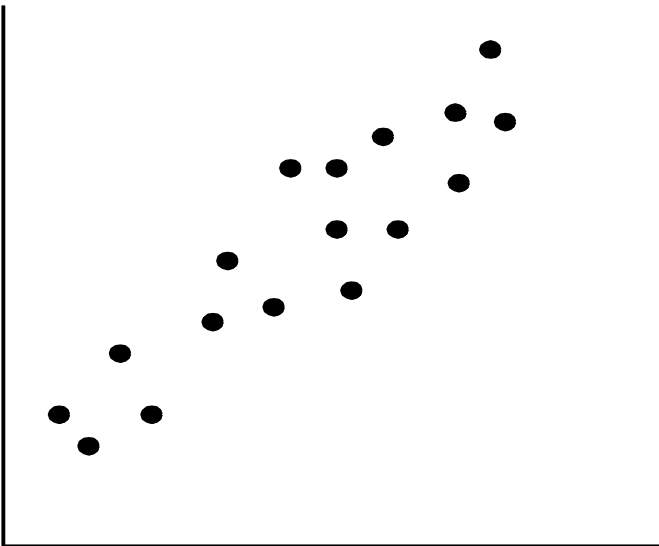
- Provides a first look at bivariate data to see clusters of points, outliers, etc
- Each pair of values is treated as a pair of coordinates and plotted as points in the plane



3.2 Descriptive Statistics Measure

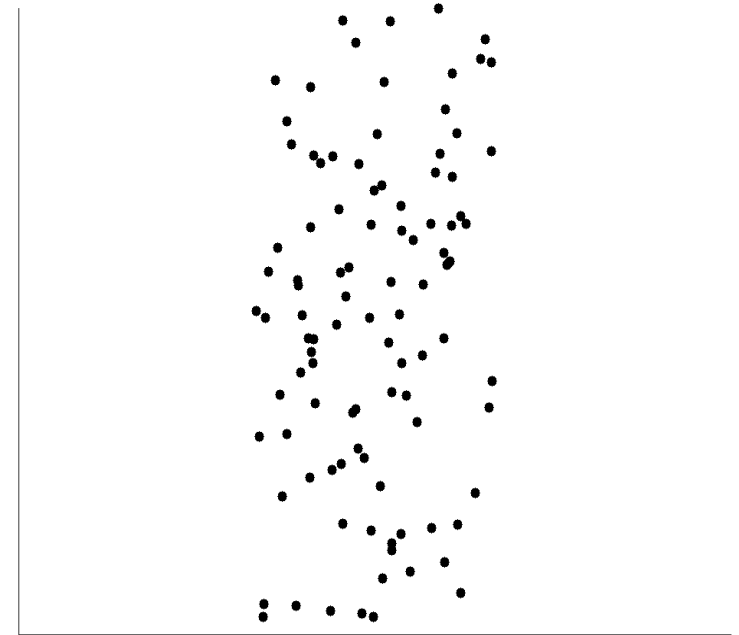
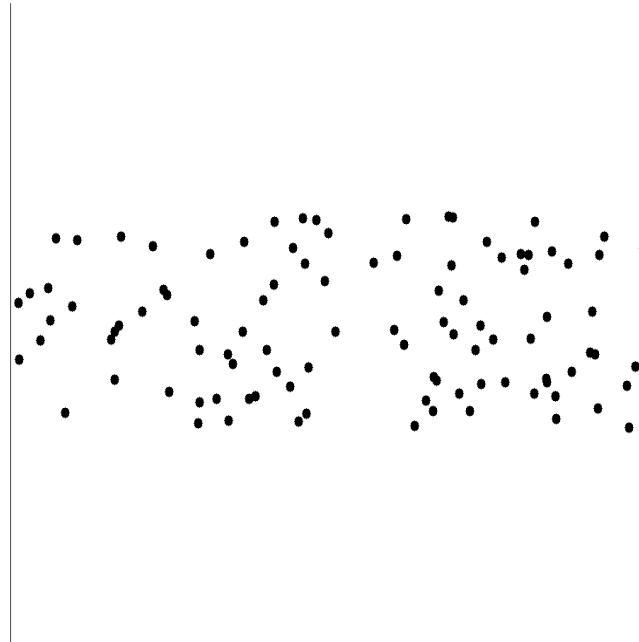
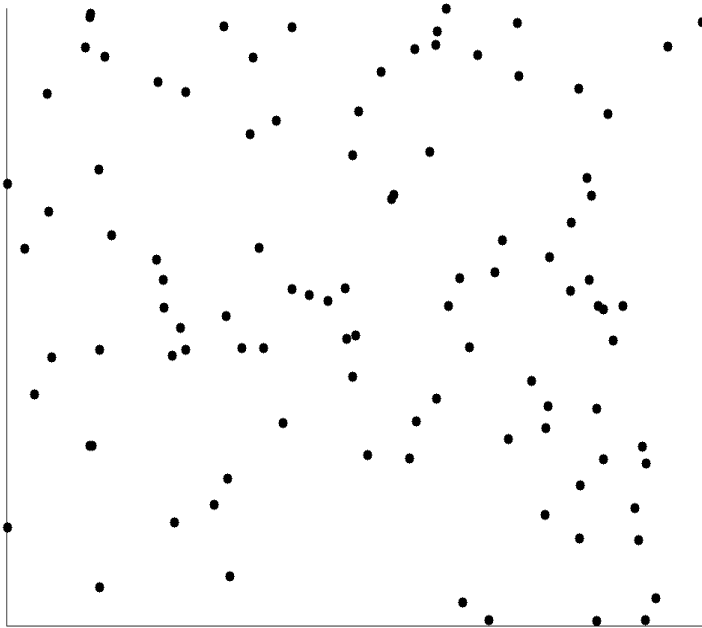
Positively and Negatively Correlated Data

- The left half fragment is positively correlated
- The right half is negative correlated



3.2 Descriptive Statistics Measure

Uncorrelated Data



Practice 2

- 1 Seats per store are 70, 62, 58, 54, 48, 44, 40, 36, 32, 30, 24, 16. Compute the mean, the median, the range, Q1, Q3 and the IQR by hand. Then state whether any store is an outlier under the $1.5 \times \text{IQR}$ rule.
- 2 The mode of the basket is 45,000 dong and the median is 97,000. Explain what kind of customer each number describes and give one business decision that should use the mode rather than the median.
- 3 Remove the twelve price-scale errors from the transaction file, then say which of these four numbers changes a lot and which barely moves: mean, median, standard deviation, IQR. Justify each answer before you compute it.



3.3

Data Visualization

The right chart, and what every chart hides

3.3 Data Visualization

What a chart is for

Gain insight

See structure, clusters, gaps and outliers that no summary statistic reports

Qualitative overview

Take in a large data set at a glance, before committing to a model

Search visually

Find the pattern first, then measure it. The eye is a fast filter

Communicate

Persuade a decision maker who will never read your code

ANSCOMBE'S WARNING

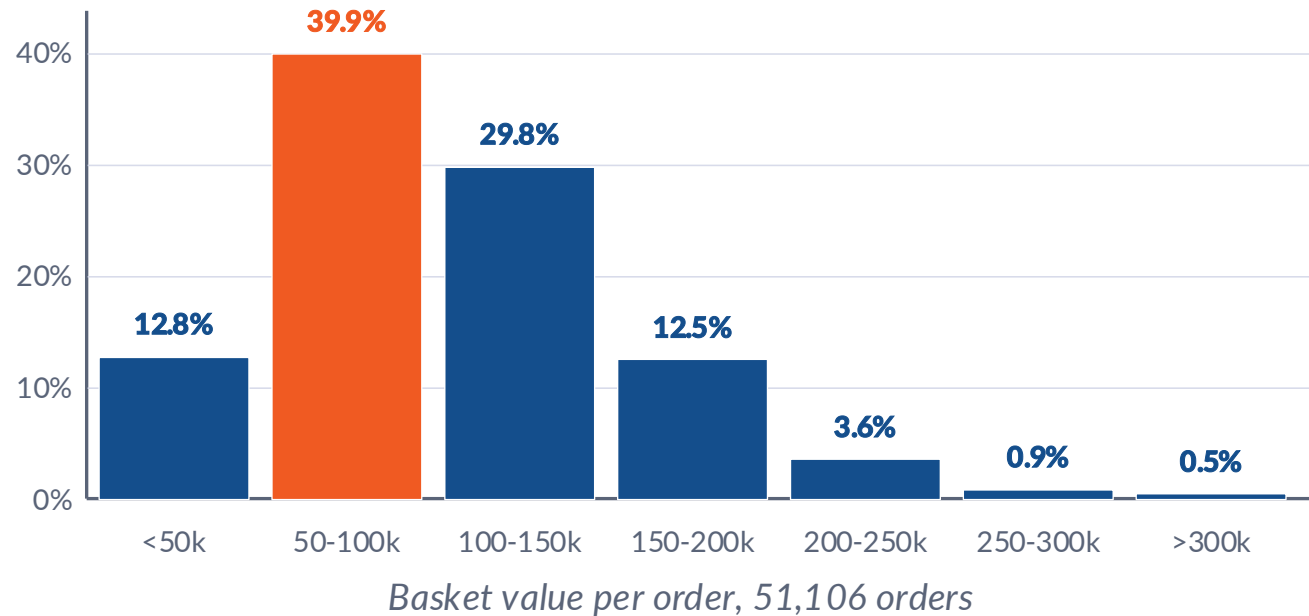
Four data sets can share the same mean, the same variance, the same correlation and the same regression line, and look nothing alike when plotted. Summary statistics compress, and compression discards. Plot the data before you trust a number computed from it.

AND THE REVERSE WARNING

A chart persuades faster than it proves. Choose the axis range, the bin width or the color scale carelessly and you will mislead an audience without intending to. Every chart in this section comes with the question: what does it hide?

3.3 Data Visualization

Histogram of the Saigon Bean basket value



How to read it

- The x axis is divided into buckets, and the height is the count or the share in each bucket
- Peak in the 50 to 100 thousand bucket, holding 39.9% of all orders
- The right tail thins out and never quite stops

BIN WIDTH IS A CHOICE

Too few bins and the shape disappears into a block. Too many and noise looks like structure. Change the bin width, and the story changes with it. Report the width you used and show one alternative when the conclusion depends on it.

3.3 Data Visualization

Histograms often tell more than boxplots

What the boxplot gives you

- Five numbers, plus the outliers, in very little space
- Several distributions side by side, which a histogram does badly
- A rule for what counts as extreme
- It cannot show a second peak, because a median and two quartiles cannot express one

What the histogram gives you

- The full shape, including two peaks, gaps and a floor
- Evidence about whether one process or two generated the data
- It depends on the bin width, which the boxplot does not
- It compares badly across many groups

THE CASE THAT DECIDES IT

Two distributions can share an identical five-number summary while one is unimodal and the other has two clear peaks. The boxplots are indistinguishable. The histograms are not. If a bimodal shape would change your decision, the boxplot alone cannot be trusted.

Use the boxplot to compare twelve stores at once. Use the histogram to understand one of them properly.

3.3 Data Visualization

Quantile plot and quantile-quantile plot

Quantile plot: one distribution

- Plot each value against the fraction of the data below it
- Shows every observation, and reads off any percentile directly
- The steeper the curve, the sparser the data in that range

Q-Q plot: two distributions

- Plot the quantiles of one distribution against those of another
- Points on the 45-degree line mean the two distributions agree
- Against a normal distribution it becomes the normality check

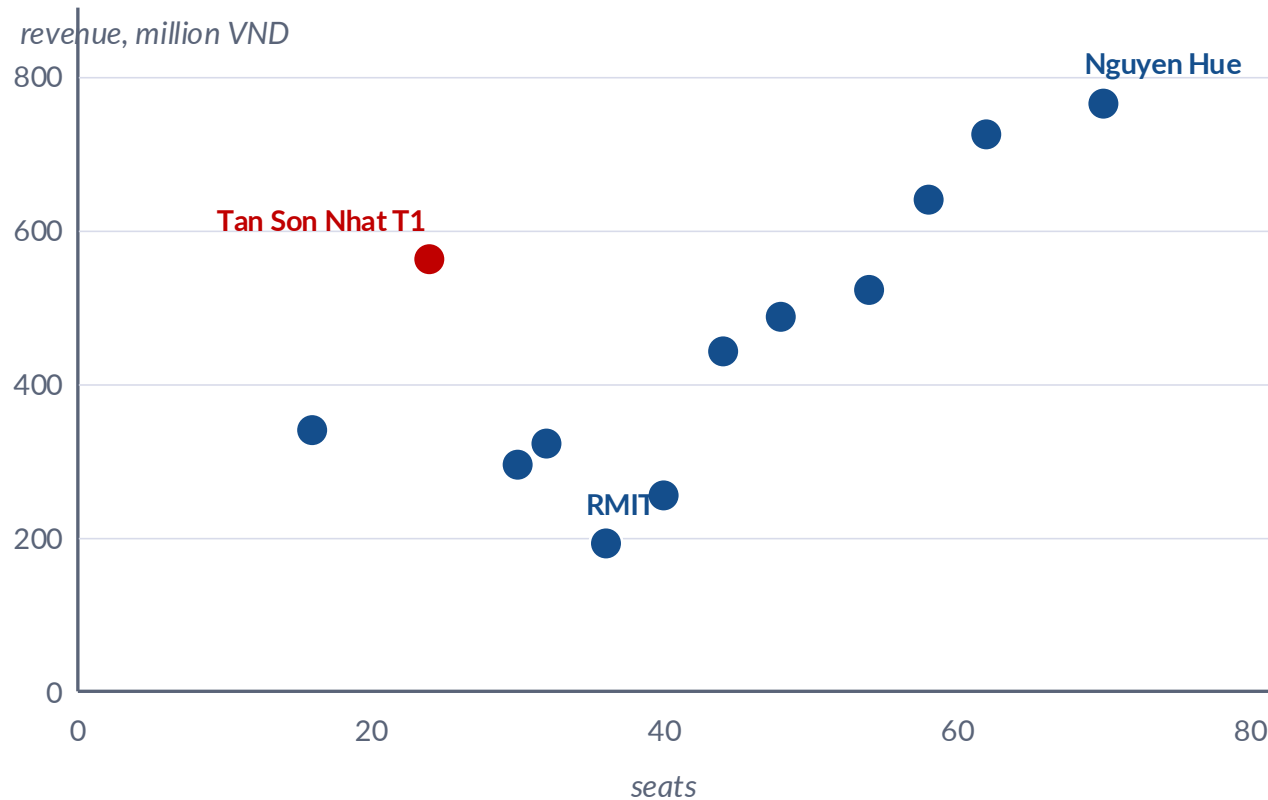
Percentile	p5	p10	p25	p50	p75	p90	p95	p99	max
Basket, VND	42,000	45,000	70,500	97,000	138,500	174,000	200,000	263,000	11,097,000

READ THE TAIL, NOT THE MIDDLE

From p5 to p95 the basket grows from 42,000 to 200,000, a factor of five. From p99 to the maximum it grows from 263,000 to 11,097,000, a factor of forty-two. The whole problem in this variable lives in the last one per cent.

3.3 Data Visualization

Scatter plot: seats against revenue, twelve stores



What the scatter shows

- Correlation of seats with revenue is 0.725, so larger stores do earn more
- The relationship is real and it is far from tight

THE POINT THAT BREAKS THE LINE

Tan Son Nhat T1 has the second fewest seats, 24, and earns 563 million dong, more than four stores with twice the seating. It is an airport unit where nobody sits down. A correlation coefficient would have hidden this. The scatter plot puts it on the screen in one second.

3.3 Data Visualization

Beyond two attributes

Geometric projection

Scatterplot matrices, parallel coordinates, landscapes.
Project many dimensions into views a person can read.

Pixel-oriented

One pixel per value, one window per attribute, color carries the magnitude. Built for very large tables.

Icon-based

Chernoff faces and stick figures. Each object becomes one icon, each attribute controls one feature of it.

Hierarchical

Tree-maps, dimensional stacking, worlds-within-worlds.
Partition the attributes into nested subspaces.

Practice 3

- 1 Using the nine percentile values on slide 26, sketch the quantile plot of the basket value by hand. Mark the median and the interquartile range on your sketch, then say in one sentence what the shape of the curve tells you.
- 2 A regional manager sees the scatter plot and concludes that adding seats raises revenue. Give two reasons from this section why that conclusion does not follow, and name one additional attribute you would plot before answering.
- 3 Choose the better chart for each question and justify it in one line. (a) Do the twelve stores differ in basket value? (b) Does the basket value have one peak or two? (c) Is the basket value normally distributed? (d) Which hour of the day is busiest?

4.4

Measuring Data Similarities and Differences

The measures that Chapter 5 is entirely built on

4.4 Similarities and Differences

Three words that get confused

Similarity

A numerical measure of how alike two objects are. Higher means more alike. Often scaled to the range $[0, 1]$.

Dissimilarity

A numerical measure of how different two objects are. Lower means more alike. Zero when the objects coincide.

Proximity

The umbrella term covering both. A proximity matrix is what clustering algorithms actually consume.

Why this section exists

- Clustering groups objects that are close, so it needs a definition of close
- k-nearest-neighbor classification needs the same definition
- Outlier detection asks which object is far from everything else
- Recommendation asks which customer resembles which

THE DECISION NOBODY DOCUMENTS

Choosing the measure is a modeling decision with consequences at least as large as choosing the algorithm. Two analysts using the same clustering method and different distance measures will report different segments and both will be defensible. Write down which measure you used and why.

4.4 Similarities and Differences

Data matrix and dissimilarity matrix

Data matrix: n objects by p attributes

- Two modes, because rows and columns mean different things
- This is the shape of every CSV file you will open

Dissimilarity matrix: n by n

- One mode, because rows and columns are the same objects
- Symmetric, with a diagonal of zeros, so only the lower triangle is stored

Data matrix (channel share, % of orders)

	in-store	app	delivery
Nguyen Hue NgH	61.5	22.3	16.2
Univ Village UVi	37.2	53.4	9.4
Tan Son Nhat TSN	91.8	6.1	2.1
Phu Nhuan PNh	46.0	34.6	19.5

Dissimilarity matrix (Euclidean)

	NgH	UVi	TSN	PNh
NgH	0			
UVi	40.05	0		
TSN	37.14	72.61	0	
PNh	20.06	23.08	56.68	0

4.4 Similarities and Differences

Proximity for nominal attributes

$$d(i, j) = (p - m) / p$$

`p` = number of attributes compared

`m` = number on which `i` and `j` match

The alternative

- Create one asymmetric binary attribute per state, then use the binary measures on the next slide

Store pair	district	store_type	m / p	d(i, j)
Nguyen Hue vs Ben Thanh	District 1 = District 1	mall = mall	2 / 2	0.00
Nguyen Hue vs Landmark 81	District 1 ≠ Binh Thanh	mall = mall	1 / 2	0.50
Nguyen Hue vs Univ. Village	District 1 ≠ Thu Duc	mall ≠ campus	0 / 2	1.00
RMIT vs Crescent Mall	District 7 = District 7	campus ≠ mall	1 / 2	0.50

4.4 Similarities and Differences

Proximity for binary attributes

		object j		
		1	0	sum
object i	1	q	r	q+r
	0	s	t	s+t
sum		q+s	r+t	p

symmetric $d = (r + s) / (q + r + s + t)$

asymmetric $d = (r + s) / (q + r + s)$

Jaccard sim = $q / (q + r + s)$

Which formula, and when

- Symmetric when a 0 is as informative as a 1, for example a yes-or-no survey answer
- Asymmetric when 1 is rare and informative, and a shared 0 means nothing
- The count t of shared zeros is simply dropped in the asymmetric case
- Jaccard similarity is one minus the asymmetric distance

JACCARD ON TWO REAL SAIGON BEAN CUSTOMERS

Customer 10001 has bought products {101, 102, 104, 105, 106, 201, 203, 301, 302, 304}. Customer 10002 has bought {101, 102, 104, 105, 106, 107, 201, 301, 302, 303}. They share eight products, and between them they have bought twelve distinct ones. Jaccard similarity = $8 / 12 = 0.667$, so the dissimilarity is 0.333. The seventeen-minus-twelve products that neither has ever bought contribute nothing, which is exactly right: two customers are not alike because both of them ignore the ceramic tumbler.

4.4 Similarities and Differences

Standardizing numeric data before you measure

Why it is not optional

- Distance is dominated by whichever attribute has the largest numeric range
- seats runs 16 to 70, revenue runs 194,000,000 to 767,000,000
- Without standardizing, revenue decides every distance and seats contributes nothing

MEAN ABSOLUTE DEVIATION IS MORE ROBUST

It does not square the deviations, so a single extreme value moves it far less than it moves the standard deviation. With twelve price-scale errors in the file, that robustness is not theoretical.

z-score	$z = (x - \mu) / \sigma$
mean absolute deviation	$s_e = (1/n) \sum x_i - m_e $ $z = (x - m_e) / s_e$
min-max	$x' = (x - \min) / (\max - \min)$

CHOOSING THE SHARE INSTEAD

Our store vectors use shares that already sum to 100, which removes the scale problem at the source. Converting to a share is often cleaner than standardizing after the fact.

4.4 Similarities and Differences

Minkowski distance and its three famous cases

$$d(i,j) = (\sum |x_{if} - x_{jf}|^h)^{1/h} \quad h \geq 1$$

$h = 1$ Manhattan (city block, L_1)
 $h = 2$ Euclidean (straight line, L_2)
 $h \rightarrow \infty$ Supremum (largest gap, L_∞)

Properties every metric must satisfy

- Non-negativity, and identity of indiscernibles: $d(i,j) = 0$ only when $i = j$
- Symmetry: $d(i,j) = d(j,i)$. Triangle inequality: $d(i,k) \leq d(i,j) + d(j,k)$

Store pair, channel-share vector	Manhattan	Euclidean	Supremum	Cosine
Nguyen Hue vs Ben Thanh, both mall	0.6	0.37	0.3	1.0000
Univ. Village vs RMIT, both campus	1.0	0.65	0.5	1.0000
Tan Son Nhat vs Dien Bien Phu, both transit	3.1	1.93	1.5	0.9999
Nguyen Hue vs Phu Nhuan, mall vs residential	31.1	20.06	15.5	0.9562
Nguyen Hue vs Univ. Village, mall vs campus	62.2	40.05	31.1	0.8193
Univ. Village vs Tan Son Nhat, campus vs transit	109.2	72.61	54.6	0.6214

All four measures agree on the ordering here, and they disagree on the scale. Stores of the same type sit under 3.2 apart on Manhattan distance, and stores of different types sit above 31.

4.4 Similarities and Differences

Proximity for ordinal attributes

Three steps, always in this order

- Replace each value by its rank r , running from 1 to M
- Map the rank to $[0, 1]$ with $z = (r - 1) / (M - 1)$
- Treat z as numeric and apply any of the distance measures

```
seats_band  rank r  z = (r-1) / (M-1)

small       1      0.0
medium      2      0.5
large       3      1.0
                M = 3
```

Store	seats	seats_band	z	d(z) to Nguyen Hue
Nguyen Hue	70	large	1.0	0.0
Van Hanh Mall	44	medium	0.5	0.5
Tan Son Nhat T1	24	small	0.0	1.0

4.4 Similarities and Differences

Cosine similarity, and when it beats distance

$$\cos(d_1, d_2) = (d_1 \cdot d_2) / (\|d_1\| \times \|d_2\|)$$

1 identical direction
0 no shared component

What it measures

- The angle between two vectors, ignoring their length entirely
- Built for sparse high-dimensional data such as term-frequency vectors and baskets

THE CASE THAT SHOWS THE DIFFERENCE

A store selling 200 coffees and 100 pastries and a store selling 20 coffees and 10 pastries have a large Euclidean distance and a cosine similarity of exactly 1. One is ten times the size of the other, and the two have the identical product mix. Which answer you want depends on the question. For segmentation by behavior, use cosine. For capacity planning, use Euclidean.

IN THE TABLE ON SLIDE 36

Nguyen Hue against Tan Son Nhat has Euclidean distance 37.14 and cosine similarity 0.9377. Nguyen Hue against University Village has a similar Euclidean distance, 40.05, and a much lower cosine, 0.8193. The two pairs are equally far apart in magnitude and they are not equally far apart in direction.

2.4 Similarities and Differences

Attributes of mixed type

$$d(i,j) = \sum_x \delta_{ij}^{(f)} d_{ij}^{(f)} / \sum_x \delta_{ij}^{(f)}$$

$\delta = 0$ when a value is missing, or when the attribute is asymmetric binary and both are 0
 $\delta = 1$ otherwise

Each type keeps its own d

- Nominal and binary: 0 if the values match, 1 if they do not
- Numeric: the normalized absolute gap. Ordinal: rank, then map to [0, 1]

Attribute	Type	Nguyen Hue	Phu Nhuan Corner	d
store_type	Nominal	mall	residential	1.00
seats_band	Ordinal	large (z = 1.0)	small (z = 0.0)	1.00
app share	Numeric, normalized	22.3%	34.6%	0.25
sells merchandise	Asymmetric binary	yes	yes	0.00

4.4 Similarities and Differences

The attributes you choose decide who counts as similar

Measured on product mix

- Every one of the twelve stores sells roughly 55% coffee, 18% tea, 25% pastry
- Nguyen Hue against University Village: Manhattan 7.7, cosine 0.9975
- Conclusion: the twelve stores are practically identical

Measured on channel mix

- In-store share runs from 36.8% at RMIT to 93.3% at Dien Bien Phu Hub
- Nguyen Hue against University Village: Manhattan 62.2, cosine 0.8193
- Conclusion: four clearly separate kinds of store

SAME TWELVE STORES, SAME FILE, OPPOSITE ANSWERS

Nothing about the data changed between the two panels. Only the attributes fed into the distance changed. A clustering run on product mix would return one undifferentiated blob. The same algorithm on channel mix returns mall, campus, residential and transit, which is the business reality.

Practice 4

- 1 Compute the Manhattan and the Euclidean distance between Ben Thanh (61.6, 22.5, 15.9) and Binh Thanh Garden (43.1, 36.8, 20.0) on the channel-share vector. Then say which of the two measures you would report to a manager, and why.
- 2 Customer A has bought {101, 102, 301} and customer B has bought {101, 105, 301, 302}. Compute the Jaccard similarity. Then explain in one sentence why the symmetric binary measure would give a misleadingly high similarity here.
- 3 A colleague clusters the twelve stores on seats alone and reports three segments. Using slide 40, give the strongest objection to that analysis, and name the two attributes you would add.

Quick check

1. store_id holds the integers 1 to 12. Which type is it?

- A. Numeric, ratio, because it is a number
- B. Nominal, because the integer is a label
- C. Ordinal, because 1 comes before 2
- D. Interval, because gaps are equal

3. Two customers share 8 products, and 12 distinct products between them. Jaccard similarity is

- A. $8 / 20 = 0.40$
- B. $8 / 12 = 0.667$
- C. $12 / 8 = 1.5$
- D. It cannot be computed without the total catalogue

2. Mode 45,000, median 97,000, mean 108,856. What shape is it?

- A. Symmetric
- B. Negatively skewed
- C. Positively skewed, with a long right tail
- D. Not enough information to say

4. Two stores have identical product mix and one is ten times larger. Which is true?

- A. Cosine similarity near 1, Euclidean distance large
- B. Both measures report them as identical
- C. Both measures report them as very different
- D. Cosine near 0, Euclidean near 0

Lab 3 in Google Colab and submission

Task	What to produce
Task 1	Load the three files, join transactions to stores and to products, and report the shape of the result
Task 2	Build the attribute-type table for all columns, then find and repair the five faults from slide 12. Keep the missing customer_id rows and add a flag instead
Task 3	Reproduce every statistic on slides 16 and 19, then draw the histogram, the boxplot and the seats-against-revenue scatter
Task 4	Practice 1 to Practice 4, one Markdown cell per exercise
Task 5	Build the 12 by 12 dissimilarity matrix of stores on the channel-share vector, and name the store pairs that are closest and furthest apart

SUBMISSION

File named StudentID_FullName_DM_Lab02.ipynb on the LMS, plus the Colab share link set to view. Runtime then Run all before you submit, so the notebook reproduces from a clean state.

Summary

- 1 A data object is a row and an attribute is a column. Attributes are nominal, binary, ordinal or numeric, and numeric splits into interval and ratio.
- 2 The type decides what you may compute. Four of the ten columns in the transaction file hold integers that must never be averaged.
- 3 Central tendency is the mean, the median and the mode. Read all three together, because their order reveals the skew.
- 4 Dispersion is the range, the quartiles, the IQR, the variance and the standard deviation. Quantile measures survive dirty data, moment measures do not.
- 5 The five-number summary drives the boxplot, and the $1.5 \times \text{IQR}$ rule flags candidates for inspection rather than rows for deletion.
- 6 Choose the chart for the question: boxplots to compare groups, histogram for shape, Q-Q for normality, scatter for a relationship between two attributes.
- 7 Proximity measures differ by attribute type: simple matching for nominal, Jaccard for asymmetric binary, Minkowski for numeric, rank mapping for ordinal, cosine when magnitude is nuisance.
- 8 Which attributes you feed into the measure decides who counts as similar. That choice belongs to the analyst and it must be written down.



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THANK YOU!

Next: Chapter 3, Basic Data Processing and Data Warehousing

